Case Study II:
Styrian Health Data

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Questions

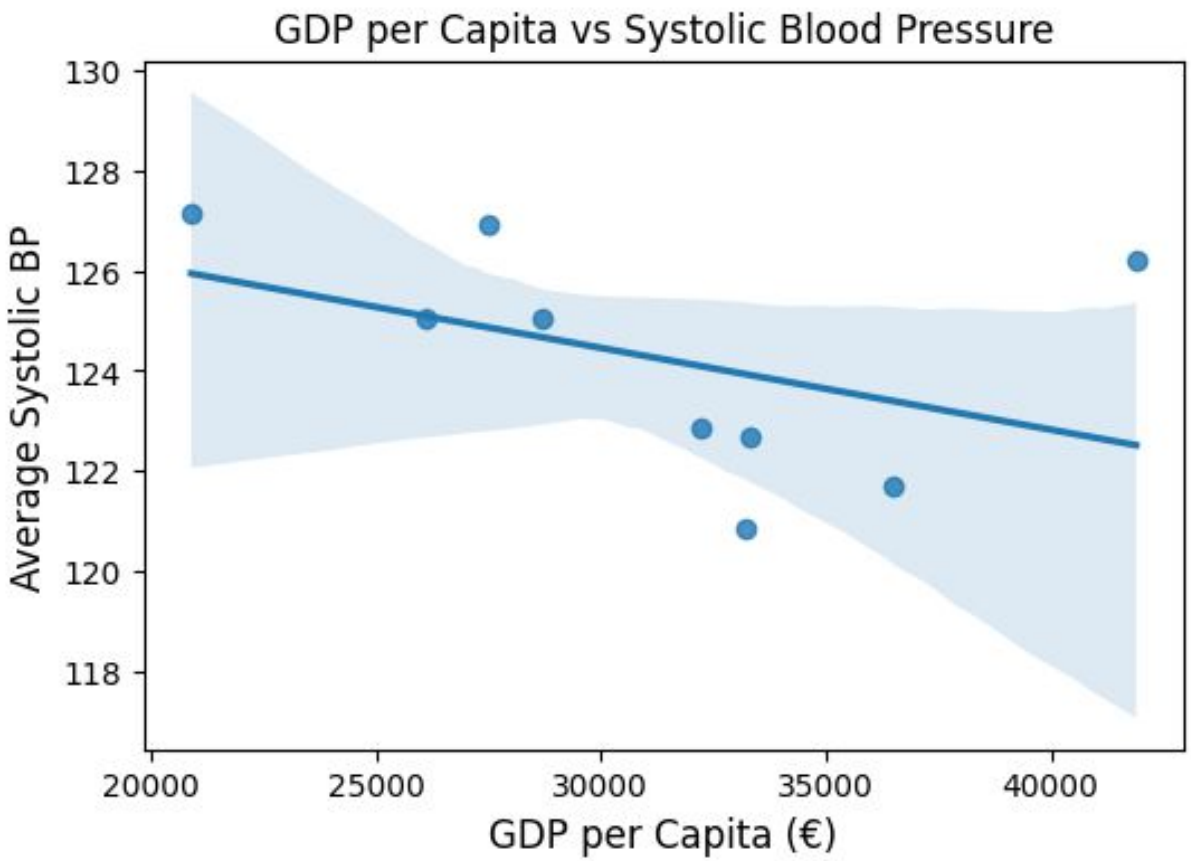
1. Does GDP per capita have an effect on BP?
2. Effects of healthystyle
3. Neighborhood effect averaging problem

Open Source Data

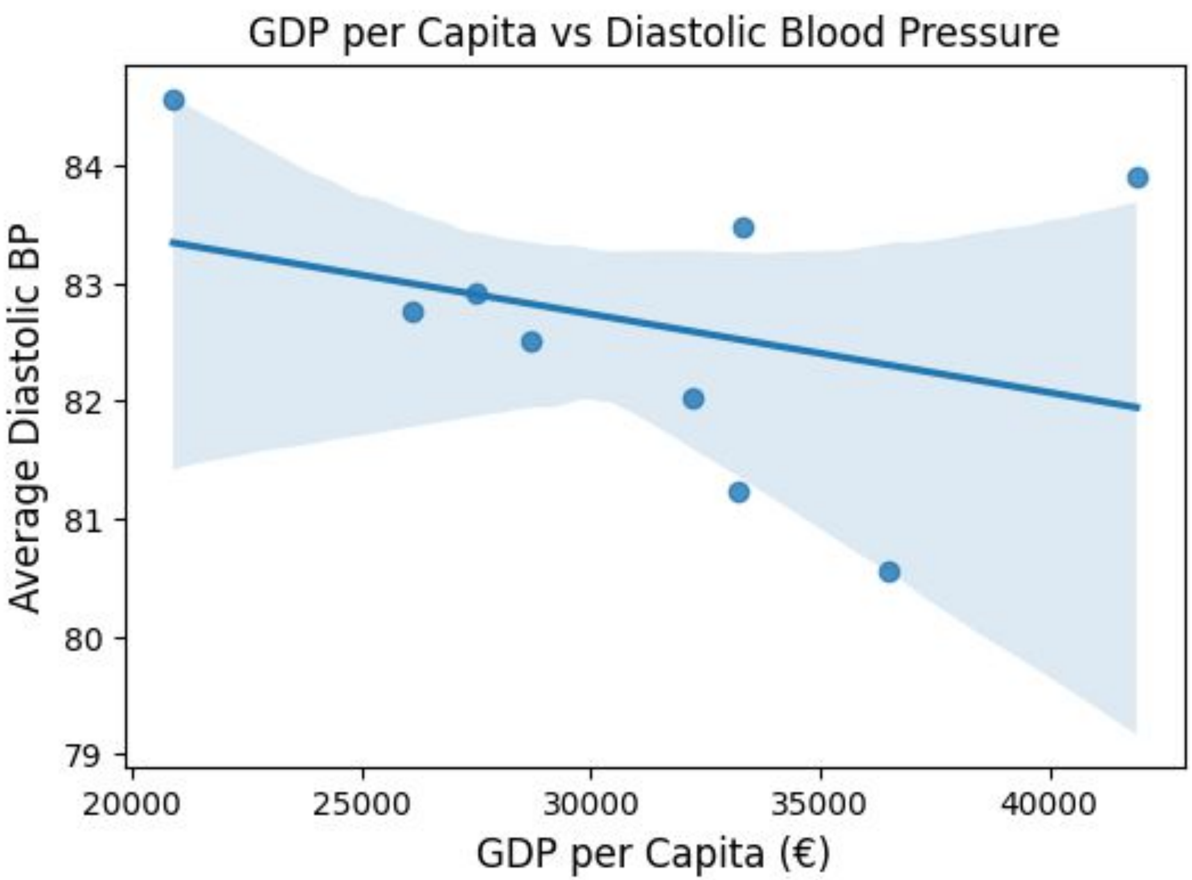
- Access: https://www.statistik.at/atlas/
  - Gross Domestic Product per Capita
  - Disposable Income per Capita

- BMI and Overweight rate
  https://link.springer.com/article/10.1007/s00508-021-01941-9

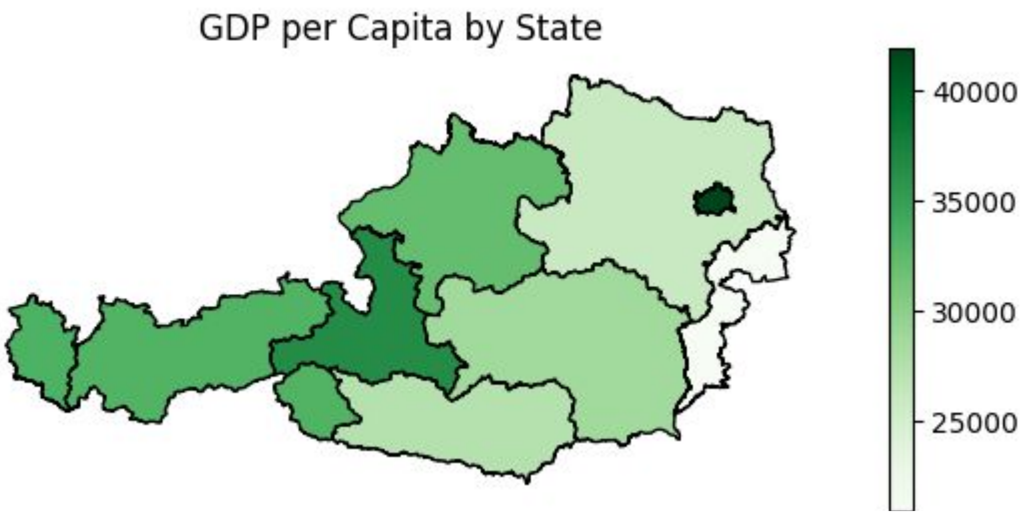
Analysis 1.1 (1/2)

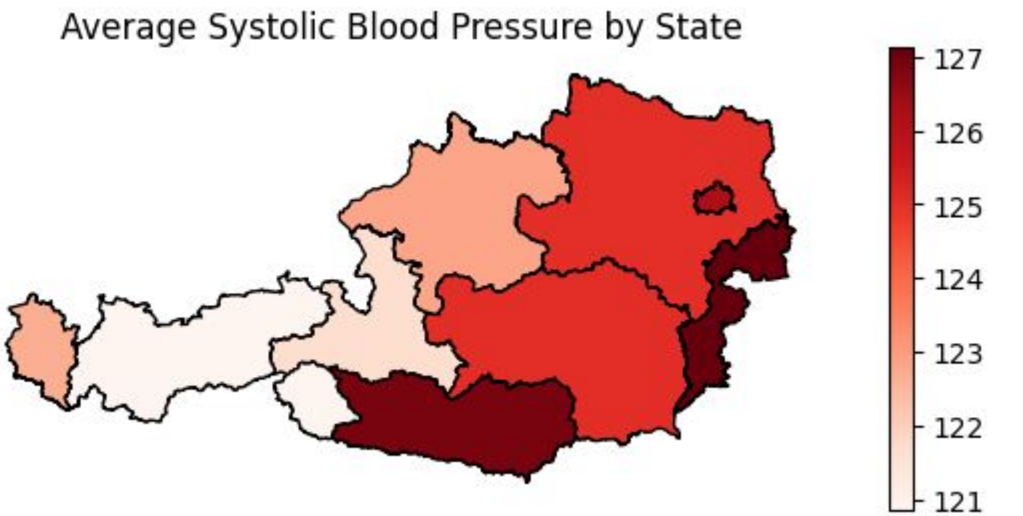


Analysis 1.1 (2/2)



Analysis 1.2 (1/2)





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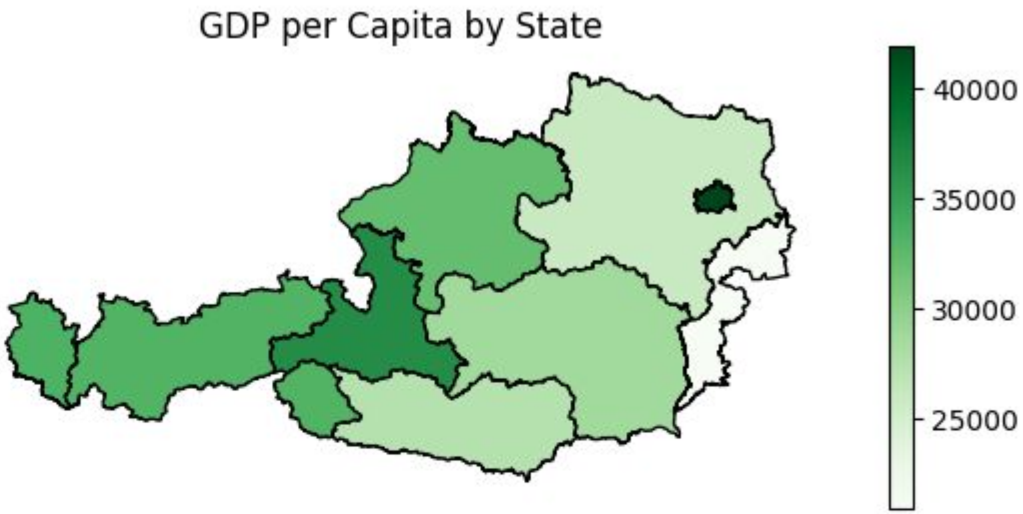
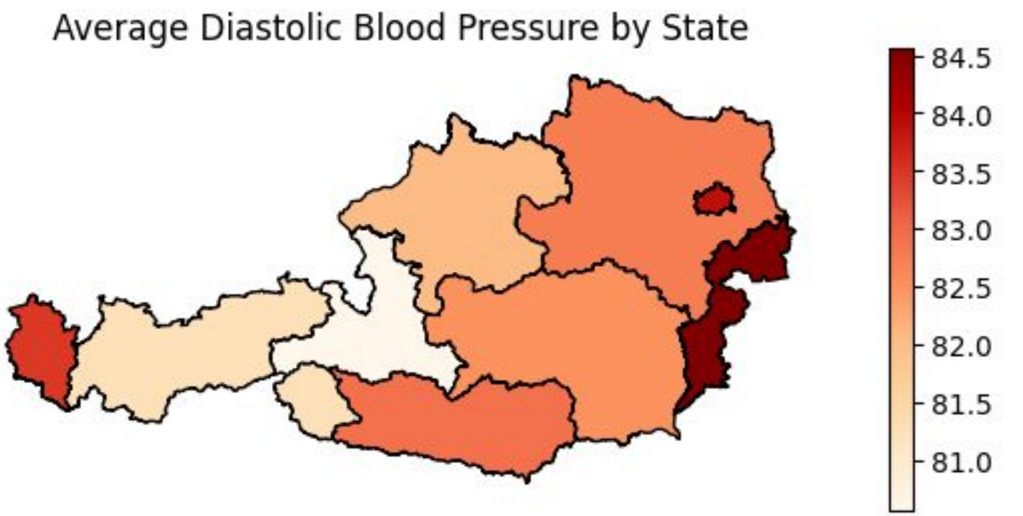
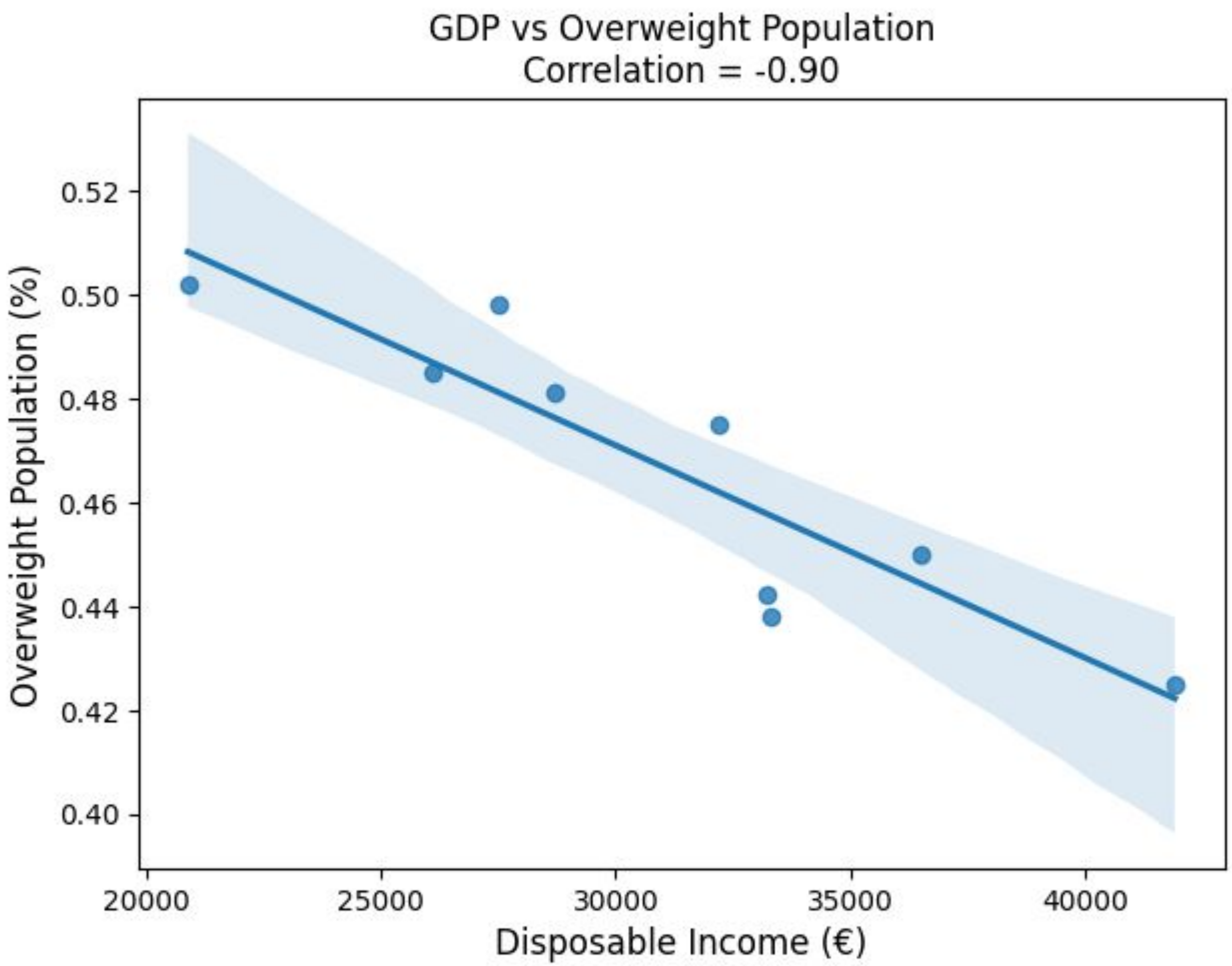
Analysis 1.2 (2/2)

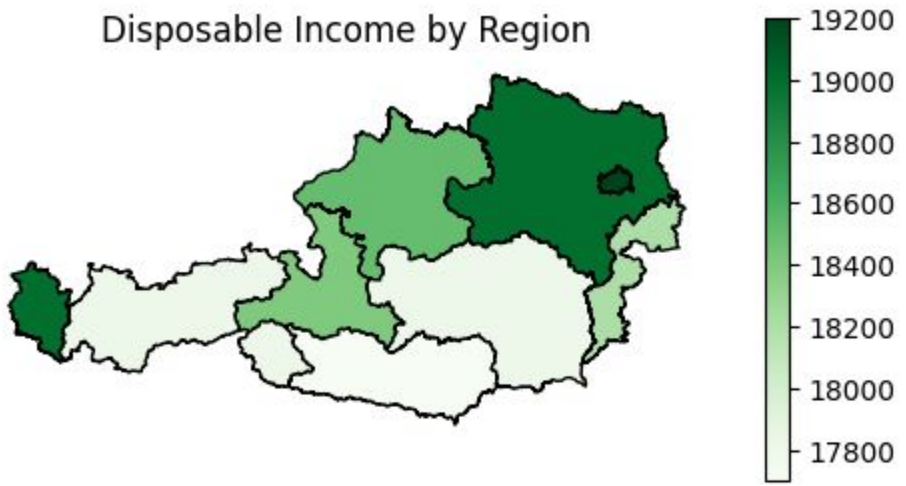


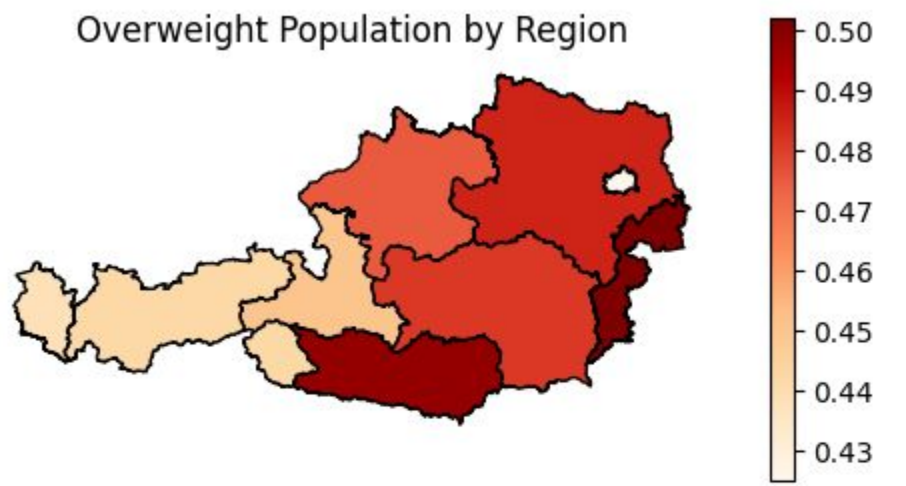


Analysis 2.1


GDP vs Overweight Population
Correlation = -0.90
Overweight Population (%)
0.52
0.50
0.48
0.46
0.44
0.42
0.40
20000
25000
30000
35000
40000
Disposable Income (€)

Analysis 2.2


Disposable Income by Region
19200
19000
18800
18600
18400
18200
18000
17800


Overweight Population by Region
0.50
0.49
0.48
0.47
0.46
0.45
0.44
0.43

NEAP

- → Analysis on the postal code level.
- → Filter out the neighbourwoods with less than 10 instances of postal codes, resulting are the 13k records.
- → Now, compute the average BP (Systolic) in the neighbourhood
- → Next, create a detrended value by subtracting individual bp value with that of the neighbourhood value

- → Model:
  - ◆ Y = X * β + Neighborhood effect + ε

NEAP

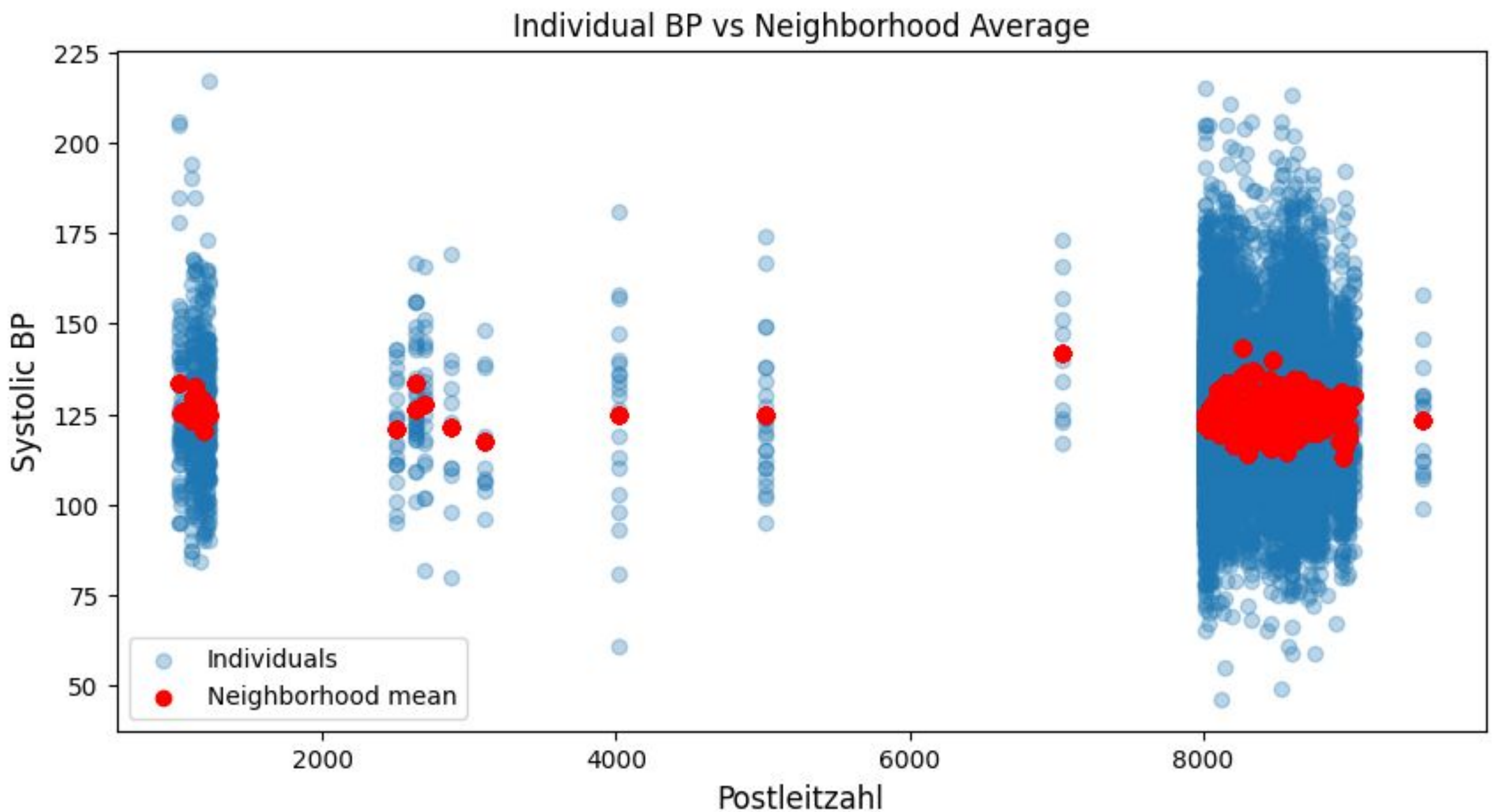
→ Model 1 (no neighbourhood effect) coefficients

const	883.42
geburtsjahr	-0.39
raucher	0.28
cholesterin_bekannt	0.34

→ Model 2 (with neighbourhood effect) coefficients

const	746.76
geburtsjahr	-0.37
raucher	0.49
cholesterin_bekannt	0.55
neighborhood_mean_bp	0.83

NEAP


Individual BP vs Neighborhood Average
225
200
175
150
125
100
75
50
Systolic BP
2000
4000
6000
8000
Postleitzahl
Individuals
Neighborhood mean

Thank you and Questions?